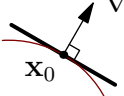


$$\{\mathbf{x} | (\mathbf{x} - \mathbf{x}_0)^T \nabla f(\mathbf{x}_0) = 0\}$$


The diagram illustrates the geometric interpretation of the equation $\{\mathbf{x} | (\mathbf{x} - \mathbf{x}_0)^T \nabla f(\mathbf{x}_0) = 0\}$. It shows a red curve representing the level set of a function f passing through the point $\mathbf{x}_0$. A thick black line is tangent to the curve at $\mathbf{x}_0$. A vector labeled $\nabla f(\mathbf{x}_0)$ points away from the level set, perpendicular to the tangent line. A right-angle symbol indicates the orthogonality between the tangent line and the gradient vector.